HK IMO Training 2021-2022
Problem Set 6
For 6 November 2021

Problem 21. In $\triangle ABC$, $\angle C = 90°$. A point D lies inside $\triangle ABC$. The lines AD, BD and CD intersect BC, CA and AB at X, Y and Z respectively. Let M be the midpoint of XY. If XZ bisects $\angle BZC$, prove that $MZ = MC$.

Problem 22. Find all positive integers n for which there exists a permutation $a_1, a_2, \ldots, a_n$ of the numbers $1, 2, 3, \ldots, n$ such that the remainders of

$$0, a_1, a_1 + a_2, \ldots, a_1 + a_2 + \cdots + a_n$$

when divided by $n + 1$ are pairwise distinct.

Problem 23. Let $\mathcal{P}$ be a convex polygon on the plane. Prove that for any point on the plane, we can reflect the point in the sides of $\mathcal{P}$ for a finite number of times such that the image lies on or inside $\mathcal{P}$.

Problem 24. Let $P(x)$ be a polynomial with real coefficients such that $P(P(x))$ is the square of a polynomial with real coefficients. Is it true that $P(x)$ is the square of a polynomial with real coefficients?